

System of Linear Equations

A collection of one or more linear equations with same number of unknowns $x_1, \dots, x_n$

e.g. $2x_1 - x_2 + 1.5x_3 = 8$ ①
 $x_1 - 4x_3 = -7$ ②

solutions: $(x_1 = -7, x_2 = -22, x_3 = 0)$, $(x_1 = 1, x_2 = -3, x_3 = 2)$,
solution set:

$$\begin{aligned} \textcircled{1} - 2 \times \textcircled{2} &\Rightarrow -x_2 + 1.5x_3 - (-8x_3) = 8 - (-14) \\ &\Rightarrow -x_2 + 9.5x_3 = 22 \\ &\Rightarrow x_2 = 9.5x_3 - 22 \end{aligned}$$

$$\textcircled{2} \Rightarrow x_1 = 4x_3 - 7$$

Then the solution set is the set of (x_1, x_2, x_3) such that
 $x_1 = 4x_3 - 7$ and $x_2 = 9.5x_3 - 22 \Rightarrow$ infinitely many solutions

A system of equations can have

{	1. no solution	inconsistent
	2. exactly one solution	consistent
	3. infinitely many solution	

Row Reduction

$$\begin{cases} X_1 + 2X_2 - X_3 = 1 \\ -X_1 + X_3 = -2 \\ X_2 + 3X_3 = 0 \end{cases} \longrightarrow \begin{array}{ccc|c} X_1 & X_2 & X_3 & \text{constant} \\ \hline 1 & 2 & -1 & 1 \\ -1 & 0 & 1 & -2 \\ 0 & 1 & 3 & 0 \end{array}$$

↑ augmented column.

equations = # rows

unknowns (variables) = # columns.

Definition. A matrix is in *row echelon form* if:

1. All zero rows are at the bottom.
2. The first nonzero entry of a row is to the right of ~~the first nonzero entry~~ of the row above. leading "1"s
3. Below the ~~first nonzero entry~~ of a row, all entries are zero. leading "1"s

Here is a picture of a matrix in row echelon form:

$$\begin{pmatrix} \boxed{1} & * & * & * & * \\ 0 & \boxed{1} & * & * & * \\ 0 & 0 & 0 & \boxed{1} & * \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

* = any number

~~$\boxed{*}$ = any nonzero number~~

$\boxed{1}$ pivotal "1"s.

Definition. A **pivot** is the first nonzero entry of a row of a matrix in row echelon form.

Definition. A matrix is in **reduced row echelon form** if it is in row echelon form, and in addition:

~~4. Each pivot is equal to 1.~~

5. Each pivot is the only nonzero entry in its column.

$$\begin{pmatrix} \boxed{1} & * & * & * & * \\ 0 & \boxed{1} & * & * & * \\ 0 & 0 & 0 & \boxed{1} & * \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

* = any number

~~$\boxed{1}$ = any nonzero number~~
pivotal "1"s

REF

$$\begin{pmatrix} \color{red}{1} & 0 & * & 0 & * \\ 0 & \color{red}{1} & * & 0 & * \\ 0 & 0 & 0 & \color{red}{1} & * \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

* = any number

$\color{red}{1}$ = pivot

RREF

Inverse of a Matrix & Characterizations of Invertible Matrices

An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C such that

$$\underbrace{CA = I}_{\substack{\text{blue arrow} \\ A^{-1}}} \quad \text{and} \quad \underbrace{AC = I}_{\substack{\text{blue arrow} \\ A^{-1}}}$$

For $n \times n$ matrix A :

$$[A|I] \xrightarrow{\text{R.R.}} [I|A^{-1}]$$

THEOREM 4

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

THEOREM 6

- a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

- b. If A and B are $n \times n$ invertible matrices, then so is AB , and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

- c. If A is an invertible matrix, then so is A^T , and the inverse of A^T is the transpose of A^{-1} . That is,

$$(A^T)^{-1} = (A^{-1})^T$$

Introduction to Determinants

Alternating Sign Pattern:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

Example of finding A_{32} .

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

then A_{32} is obtained by crossing out row 3 and column 2,

$$\begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

so that

$$A_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$C_{32} = (-1)^{3+2} (1) \det(A_{32})$$

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\det A = \underline{a_{11} \det A_{11}} - \underline{a_{12} \det A_{12}} + \cdots + \underline{(-1)^{1+n} a_{1n} \det A_{1n}}$$

$$= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j}$$

← cofactor expansion across the first row.

$$(i,j) - \text{cofactor} = C_{ij} = (-1)^{i+j} a_{ij} \det(A_{ij})$$

Theorems:

If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

A square matrix A is invertible if and only if $\det A \neq 0$.

Properties of Determinants

Row Operations

Let A be a square matrix.

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \det A$.

If A is an $n \times n$ matrix, then $\det A^T = \det A$.

Multiplicative Property

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

Now we have two general techniques for computing determinants of $n \times n$ matrix A ($n \geq 2$)

Example:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 10 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

[Method 1]: (Cofactor Expansion)

$$\det(A) = (-1)^{2+1} (-3) \det \begin{bmatrix} 1 & 2 \\ 5 & 1 \end{bmatrix} = 3 \cdot (1 - 10) = -27$$

[Method 2]: (Row Operations)

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 10 & 0 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow[\text{change sign}]{R_1 \leftrightarrow R_2} \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow[\text{no change}]{R_3 \rightarrow R_3 - 5R_2} \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix}$$

$$\det(A) = -\det \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix} = -1((-3) \times 1 \times (-9)) = -27$$

Linear Independence & Span

Linear Combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ is $\vec{w} = \underbrace{a_1}_{\text{weights}} \vec{v}_1 + \underbrace{a_2}_{\text{weights}} \vec{v}_2 + \dots + \underbrace{a_n}_{\text{weights}} \vec{v}_n$

An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_p \mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p = \mathbf{0} \quad (2)$$

Let's say $c_1 \neq 0$ then $\mathbf{v}_1 = -\frac{c_2}{c_1} \mathbf{v}_2 + \dots + \frac{c_p}{c_1} \mathbf{v}_p$

If you can write $\mathbf{v}_1$ in a linear combination of $\{\mathbf{v}_2, \dots, \mathbf{v}_p\}$, then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly dependent.

Span

If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$ is denoted by $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated) by $\mathbf{v}_1, \dots, \mathbf{v}_p$** . That is, $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is the collection of all vectors that can be written in the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p$$

with $c_1, \dots, c_p$ scalars.

How to determine whether a set is linearly independent?

- ① $[\mathbf{v}_1 | \mathbf{v}_2 | \dots | \mathbf{v}_p]$
- ② row reduce to REF
- ③ If there are pivots in every column, then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is L.I.

The pivot columns in the original matrix are the L.I. set of vectors

e.g. $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_5$ are the pivot columns, then
 $\text{span}\{\underbrace{\mathbf{v}_1, \dots, \mathbf{v}_5}_{\text{L.I.}}\} = \text{span}\{\underbrace{\mathbf{v}_1, \dots, \mathbf{v}_p}_{\text{L.D.}}\}.$

Introduction to Linear Transformation

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

matrix of transformation
(Standard matrix)

$$\text{Matrix transformation: } T(\vec{x}) = A\vec{x}$$

Let's say $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $A: m \times n$ matrix

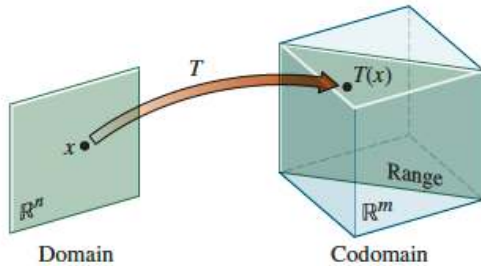


FIGURE 2 Domain, codomain, and range of $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

$$\text{e.g. } T(\vec{x}) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \vec{x}$$

domain: $\mathbb{R}^2$

codomain: $\mathbb{R}^2$

$$\text{range: } \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2 \mid a=0 \right\}$$

Find standard matrix:

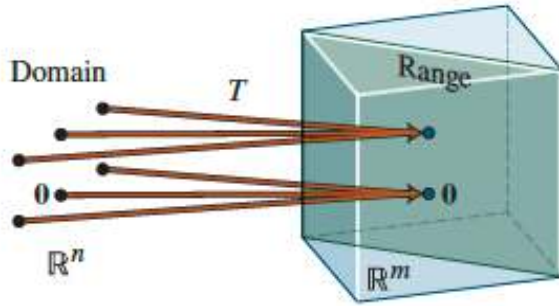
$$T(\vec{x}) = A\vec{x} \quad (T: \mathbb{R}^n \rightarrow \mathbb{R}^m)$$

$$A = \begin{bmatrix} T(\vec{e}_1) & \dots & T(\vec{e}_n) \end{bmatrix}$$

$$\text{e.g. } T: \mathbb{R}^2 \rightarrow \mathbb{R}^3 \quad T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \\ 3 & 1 \end{bmatrix} \quad T(\vec{x}) = A\vec{x}$$

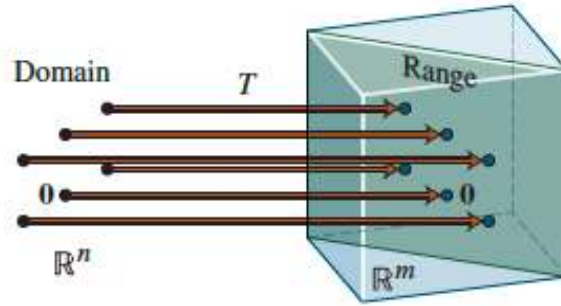
The Matrix of a Linear Transformation



T is not one-to-one

not everything in the codomain can be mapped to

$T(\vec{x}) = A\vec{x}$
 A has pivot in every row $\Leftrightarrow T$ is onto



T is one-to-one

every point in the range is mapped from only one point in the domain.

$T(\vec{x}) = A\vec{x}$
 A has pivot in every column $\Leftrightarrow T$ is one-to-one